

HALLUM, Sweden

WMO: 025420

Lat: 58.3218N

Lon: 13.0379E

Elev: 233

StdP: 14.57

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.7	8.8	-0.5	5.4	3.1	5.6	7.4	9.4	29.9	43.3	27.2	41.6	1.3	190	0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	80.4	65.4	77.0	64.0	73.6	62.5	67.5	76.4	65.9	74.2	64.0	71.3	5.5	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.5	91.7	70.7	62.7	85.9	69.1	61.0	80.7	67.2	32.0	76.2	30.7	74.4	29.4	71.5	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.8	20.0	17.6	-3.8	84.8	9.2	2.9	-10.5	86.9	-15.9	88.6	-21.1	90.3	-27.8	92.4		
(5)			-3.7	70.2	8.7	1.5	-10.0	71.3	-15.1	72.2	-20.0	73.0	-26.4	74.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.8	28.6	29.3	34.0	43.4	51.8	58.0	62.0	60.5	53.7	44.9	37.9	32.2		
	DBStd	13.64	9.16	8.51	8.14	5.96	6.22	4.99	4.81	4.41	5.03	6.48	6.73	9.13		
	HDD50	3124	662	580	497	212	51	2	0	1	22	180	364	553		
	HDD65	7431	1127	1000	962	649	410	218	122	151	339	622	813	1018		
	CDD50	1219	0	0	0	13	107	242	372	327	134	23	1	0		
	CDD65	51	0	0	0	0	1	8	29	12	1	0	0	0		
	CDH74	649	0	0	0	0	35	151	339	117	7	0	0	0		
	CDH80	95	0	0	0	0	2	20	62	11	0	0	0	0		
	WSAvg	7.2	8.2	7.5	7.4	7.1	6.9	6.4	6.0	6.6	7.1	7.1	7.6	8.1		
	PrecAvg	25.5	1.8	1.5	1.2	1.5	1.9	2.9	3.1	2.8	2.4	2.5	2.0	1.9		
(15) Precipitation	PrecMax	30.5	3.3	3.3	2.1	3.5	3.7	6.4	6.0	5.7	5.5	4.9	4.0	3.9		
	PrecMin	20.4	0.3	0.2	0.3	0.2	0.5	0.3	0.5	0.5	0.3	0.3	0.4	0.6		
	PrecStd	2.7	0.9	0.7	0.5	1.0	0.9	1.4	1.5	1.3	1.4	1.1	0.9	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.6	47.5	56.7	68.6	78.3	82.6	85.7	81.2	74.7	63.3	54.0	50.0		
		MCWB	44.4	44.6	47.2	54.5	61.1	65.5	66.9	65.8	62.5	57.5	52.1	48.6		
	2%	DB	43.9	44.2	51.4	64.0	73.5	78.7	81.2	77.3	69.2	58.4	51.4	47.4		
		MCWB	42.0	42.2	44.8	51.9	58.8	64.2	66.3	64.5	60.1	55.0	49.7	46.0		
	5%	DB	41.7	42.1	47.1	59.4	69.4	74.3	78.1	73.7	65.5	56.4	49.6	45.4		
		MCWB	40.1	40.6	42.3	49.3	57.4	62.3	65.4	63.2	58.6	54.0	48.1	43.9		
	10%	DB	39.8	40.1	43.9	54.9	64.2	70.2	74.0	70.2	62.8	54.4	47.5	43.2		
		MCWB	38.5	38.6	40.7	46.7	54.6	59.9	63.7	61.8	57.1	52.0	46.1	41.9		
	0.4%	WB	45.0	46.0	48.5	56.8	64.4	68.5	70.1	68.7	64.6	58.2	52.6	48.7		
		MCDB	46.2	47.5	54.8	67.1	74.1	77.8	78.7	75.8	70.8	61.4	53.6	49.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	42.4	42.7	45.6	52.8	61.1	65.4	68.0	66.6	61.2	56.2	50.0	46.3		
		MCDB	43.4	43.7	50.2	61.4	69.0	75.7	77.5	73.5	66.8	58.0	51.0	47.3		
	5%	WB	40.3	40.8	43.1	50.0	58.4	63.2	66.6	64.6	59.4	54.3	48.2	44.1		
		MCDB	41.4	41.9	46.1	57.9	67.0	72.6	75.7	71.1	64.5	56.1	49.4	45.3		
	10%	WB	38.5	38.6	41.1	47.7	56.1	61.0	64.9	62.9	57.7	52.4	46.2	41.9		
		MCDB	39.6	40.0	43.7	53.4	62.9	68.5	72.5	68.4	61.9	54.2	47.4	43.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.7	13.5	17.9	18.7	19.4	18.7	17.2	15.8	12.3	8.2	8.5		
		MCDBR	9.2	9.4	18.0	25.9	26.3	26.7	26.4	23.1	20.3	13.1	9.4	8.6		
		MCWBR	9.0	7.7	12.2	15.5	14.0	14.5	13.1	12.0	12.6	10.0	9.0	8.4		
		MCDBR	8.9	8.4	15.7	22.1	22.2	24.3	22.4	19.1	18.2	12.4	9.5	8.7		
	5% WB	MCWBR	8.7	7.4	11.4	13.9	12.8	14.1	12.2	11.3	11.9	10.0	9.2	8.6		
(40) Clear-Sky Solar Irradiance	taub		0.278	0.313	0.333	0.371	0.368	0.358	0.379	0.374	0.344	0.333	0.301	0.263		
	taud		2.240	2.315	2.375	2.329	2.397	2.447	2.407	2.420	2.492	2.443	2.315	2.176		
	Ebn at Noon		178	228	259	265	274	279	269	262	254	220	168	142		
	Edh at Noon		15	22	27	34	34	33	33	31	24	20	14	11		
(44) All-Sky Solar Radiation	RadAvg		110	292	770	1240	1634	1828	1638	1292	882	399	143	76		
	RadStd		20	49	75	157	141	102	209	118	106	63	27	8		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page